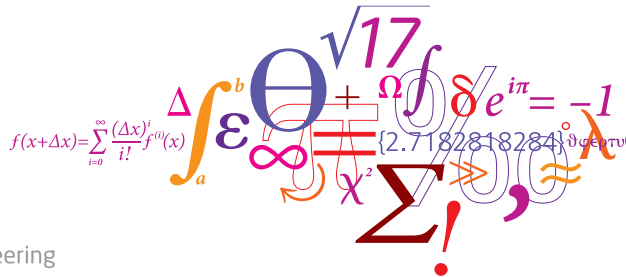
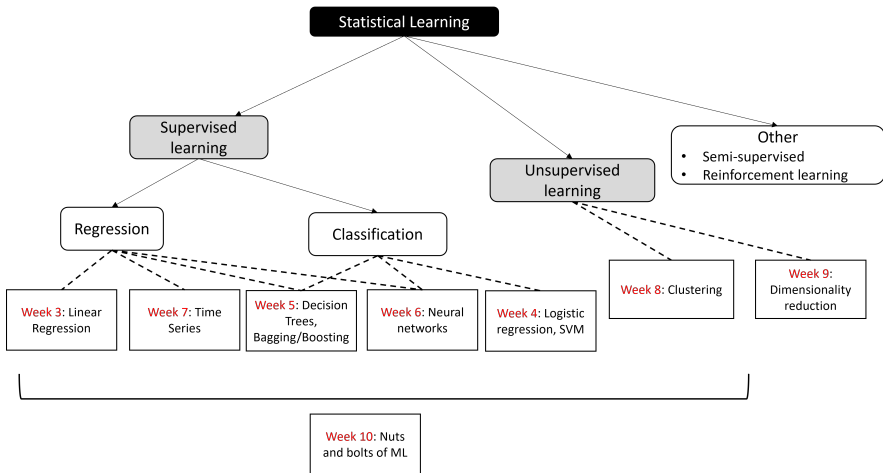


Lecture 7 - Time Series Analysis



Recap



Learning Objectives

- To explain the basic nature of time-series data and how it differs from cross-sectional data
- To explain the components of a time-series and the concept of stationarity and autocorrelation
- To analyze whether given time-series data is stationary and to handle non-stationarity through suitable transformations
- To formulate basic time-series models (ARIMA class)
- To train time-series models using *statsmodels*, make forecasts, and assess model performance

Outline

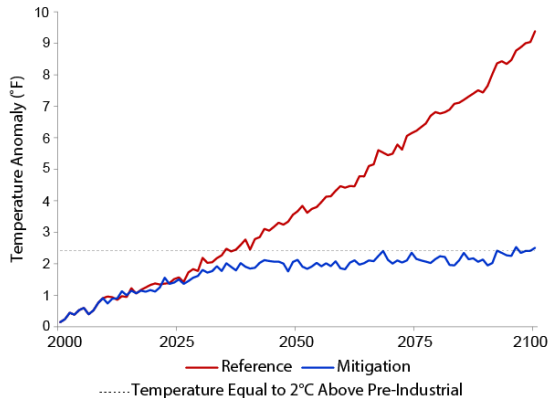
- Motivation
- Basic concepts in time series analysis
- Modeling
 - Auto Regressive (AR) models
 - Moving Average (MA) models
 - ARIMA, ARIMAX, SARIMA

Why Time Series Analysis?

- **Time series:** An ordered sequence of values of a variable at equally spaced time intervals.

Figure 1. Change in Global Mean Temperature with and without Global GHG Mitigation

Time series of global annual mean surface air temperature relative to present-day (1980-2009 mean) for IGSM-CAM under the Reference and Mitigation scenarios with a climate sensitivity (CS) of 3°C.



For more information, visit EPA's "Climate Change in the United States: Benefits of Global Action" at www.epa.gov/cira.

Why Time Series Analysis?

- Temporal dependence: observation collected at a particular time is related to or influenced by previous observation(s)

Why Time Series Analysis?

- Temporal dependence: observation collected at a particular time is related to or influenced by previous observation(s)
- Applications
 - Obtain an understanding of the underlying structure that produced the observed data
 - Fit a model and proceed to forecasting, monitoring or control

Some concepts in Time Series analysis

- Components of a time-series
- Time-series decomposition
- Stationarity
- Trend and Seasonality
- Autocorrelation

Components of a time-series

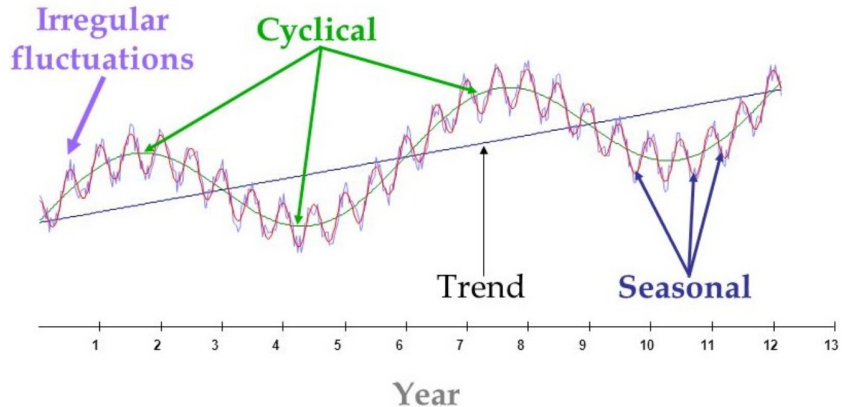


Image from: <https://www.kaggle.com/code/kushal1506/timeseries-s-p500-stockanalysis-arima-lstm>

Components of a time-series

- *Trend*: A long-term upward or downward movement in the data, indicating a general increase or decrease over time
- *Seasonality*: A repeating pattern in the data that occurs at regular intervals, such as daily, weekly, monthly, or yearly.
- *Cyclical*: A pattern in the data that repeats itself after a specific number of observations (unspecified time interval), which is not necessarily related to seasonality (e.g., economic cycles)
- *Irregularity*: Random fluctuations in the data that cannot be easily explained by trend, seasonality, or cycle.

Time-series decomposition

- Additive decomposition

$$y_t = S_t + T_t + R_t$$

where y_t is the time-series data, S_t is the seasonal component, T_t is the trend-cycle component, R_t is the remainder component

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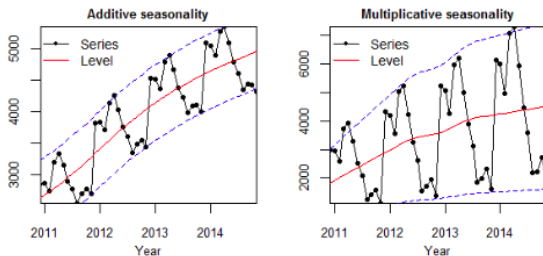
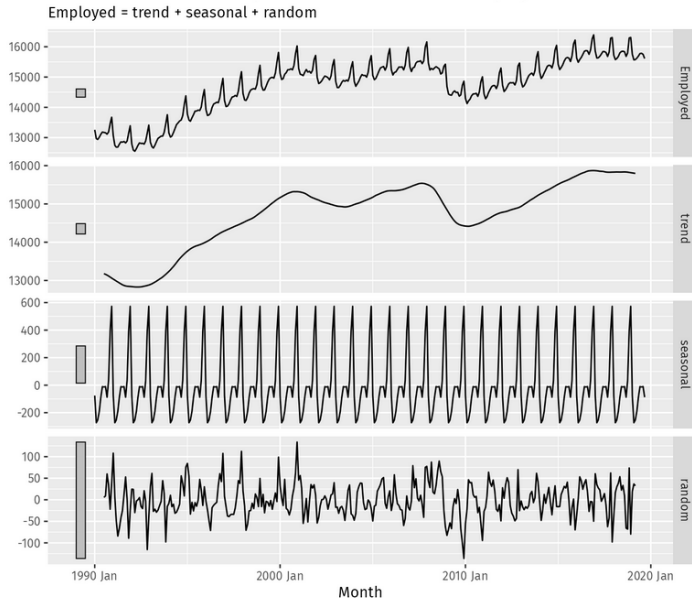


Image from: <https://kourentzes.com/forecasting/2014/11/09/additive-and-multiplicative-seasonality/>

Time-series decomposition



Total retail employment (in thousands) in the US; Image from: <https://otexts.com/fpp3/classical-decomposition.html>

Time-series decomposition

- Throughout the lecture we will use the *statsmodels* module
- Seasonal decomposition using moving averages can be done using the *statsmodels.tsa.seasonal.seasonal_decompose* function
 - Additive or multiplicative model

Time-series decomposition

- Throughout the lecture we will use the *statsmodels* module
- Seasonal decomposition using moving averages can be done using the *statsmodels.tsa.seasonal.seasonal_decompose* function
 - Additive or multiplicative model
- Moving averages is the most naive approach, more sophisticated techniques exist (STL - Seasonal and Trend decomposition using Loess, SEATS - Seasonal Extraction in ARIMA Time Series, etc.)
- Can be used to understand the nature of your time-series data, but also for forecasting

Stationarity

- Intuitively, a *stationary time series* is one whose statistical properties are constant over time

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 - *Weak stationarity*: same mean and variance at all time points, and the covariance between variables at any two time points t and $t - k$ (autocovariance) depends only on k and not the location of points in time
- Why is it important?
 - Easier to model and analyze
 - Common assumption in many practices and tools including estimation, forecasting, inference, etc.

Stationarity

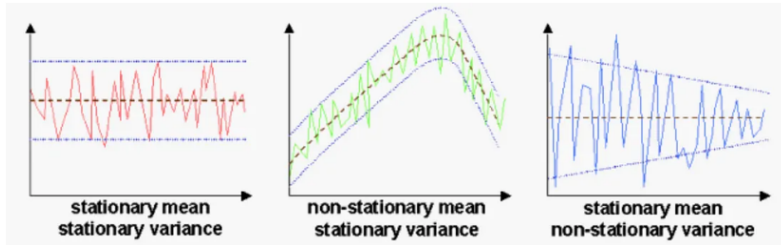


Image from: <https://towardsdatascience.com/stationarity-in-time-series-analysis-90c94f27322>

Checking for Stationarity

- Visualize (plot)
- Compare distributions in different parts (time windows) of dataset
- Augmented Dickey–Fuller (ADF) test (*statsmodels.tsa.stattools.adfuller*)
 - Null hypothesis is that a 'unit root' exists in a given time-series sample (non-stationary)
 - Alternate hypothesis is that the sample is stationary (or trend-stationary)
- Autocorrelation function (ACF)

Handling non-stationarity

- Trend adjustments (eliminate the trend)
 - Fit a regression model describing the trend component and then subtract it out of the original observations

$$E(y_t) = \beta_0 + \beta_1 t$$

$$x_t = y_t - E(y_t)$$

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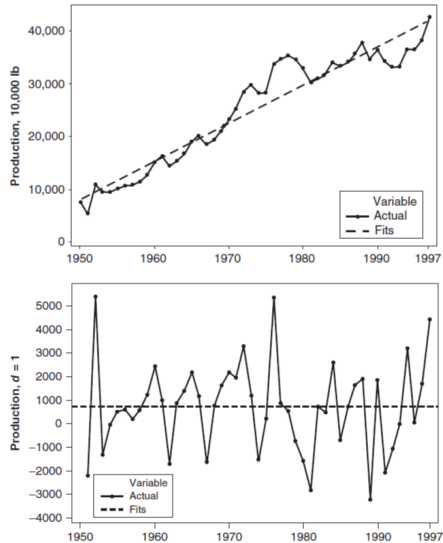
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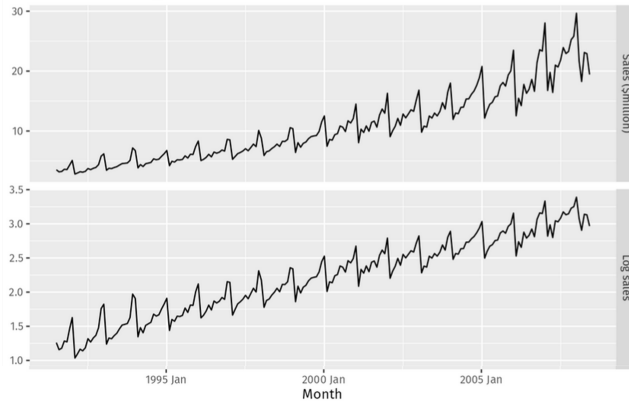
- Transformations to stabilize variance (e.g., log. power transformations)
 - Transformations + differencing
- Important: trend adjustments and transformations alone may not be sufficient to ensure stationarity if seasonal variations are present (discussed later)

Handling non-stationarity



Annual cheese production in the US; Image from: Montgomery, Douglas C., Cheryl L. Jennings, and Murat Kulahci. Introduction to time series analysis and forecasting. John Wiley & Sons, 2015.

Handling non-stationarity



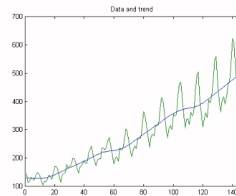
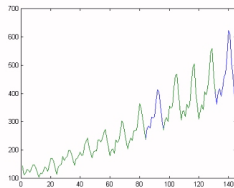
Antidiabetic sales data from Australia; Image from: <https://otexts.com/fpp2/stationarity.html>

Playtime!

- Do part 1 of "9. Time Series analysis concepts.ipynb" notebook.
- Expected duration: 30 min

Trend

- *Trend* is a general systematic linear or (most often) nonlinear component that changes over time and does not repeat
- Identifying the trend is extremely important for decision making in almost any area!
- It is a key part of a prediction



from: <http://www.es.lancs.ac.uk/cres/captain/dhrdemo.html>

Identifying and handling trends

- How to check?
 - Visually
 - Model fitting

Identifying and handling trends

- How to check?
 - Visually
 - Model fitting
- How to remove trend ("detrending")?
 - Trend adjustment: subtract fitted model from the signal ($y'_t = y_t - \hat{y}_t$)
 - Differencing ($y'_t = y_t - y_{t-1}$)

Seasonality

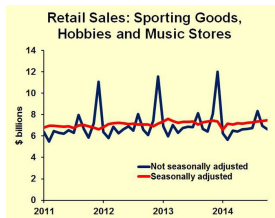
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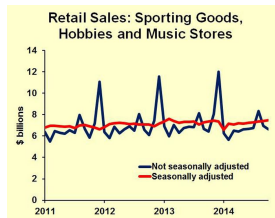
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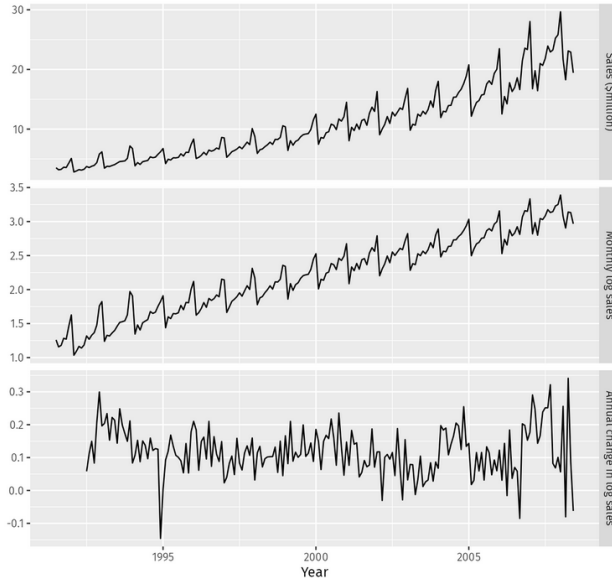


from: <https://www.forbes.com/sites/billconerly/2014/12/15/seasonality-in-business-and-economics-a-primer/>

Identifying and handling seasonality

- Checking for seasonality
 - Visualize (plot)
 - Hypothesis testing (e.g. put different dummies, for each possible seasonality, verify their importance in the model)
 - Seasonal decomposition (using moving averages), fourier series
- How to remove seasonality?
 - Seasonal adjustment
 - Intuition: each observation is adjusted by how much the season mean tends to be below or above the global (stationary) mean
 - In the simple additive decomposition model, the seasonally adjusted data are given by $y_t - S_t$
 - Seasonal differencing: If m is the seasonality, $y_t - y_{t-m}$ is the seasonally differenced data

Seasonality (example)



Antidiabetic sales data from Australia; Image from: <https://otexts.com/fpp2/stationarity.html>

Recap: Correlation and Partial Correlation

- Correlation between two random variables X and Y

$$\rho_{XY} = \frac{COV(X,Y)}{\sigma_X \sigma_Y}$$

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 - Define the errors (residuals) $X^* = X - \hat{X}$, and $Y^* = Y - \hat{Y}$
 - *Partial correlation* between X and Y after adjusting for Z is defined as the correlation between X^* and Y^* , $\rho_{X^*Y^*}$

Autocorrelation

Date	Value	Value _{t-1}	Value _{t-2}
1/1/2017	200	NA	NA
1/2/2017	220	200	NA
1/3/2017	215	220	200
1/4/2017	230	215	220
1/5/2017	235	230	215
1/6/2017	225	235	230
1/7/2017	220	225	235
1/8/2017	225	220	225
1/9/2017	240	225	220
1/10/2017	245	240	225

<https://www.business-science.io/timeseries-analysis/2017/08/30/tidy-timeseries-analysis-pt-4.html>

Autocorrelation

- Generally, *autocorrelation* is the correlation of a signal y_t with a lagged copy of itself y_{t-L} :

$$\rho(L) = \frac{\sum_{t=L}^T (y_t - \bar{y}_t)(y_{t-L} - \bar{y}_{t-L})}{\sigma_{y_t} \sigma_{y_{t-L}}}$$

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- Partial autocorrelation is the correlation where the effect of intermediate lags is removed.
 - Thus, partial autocorrelation with lag $L = h$, is given by:

$$pacf(L = h) = Corr(y_{t+h} - \hat{y}_{t+h}, y_t - \hat{y}_t)$$

- where (for a zero mean time series we don't need the intercept)

$$\hat{y}_{t+h} = \beta_1 y_{t+h-1} + \beta_2 y_{t+h-2} + \dots + \beta_{h-1} y_{t+1}$$

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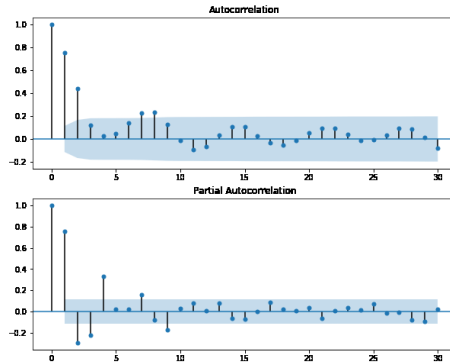
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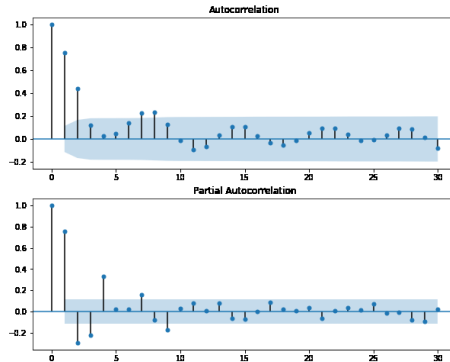
Autocorrelation

- PACF allows to see “cleaned” effects of individual variables



Autocorrelation

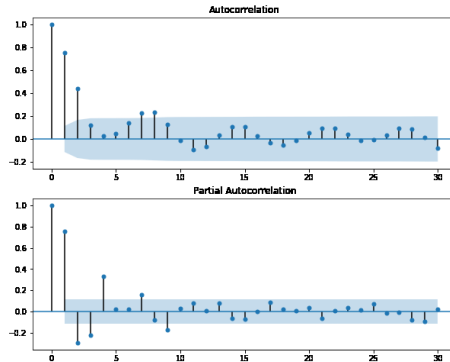
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- Shaded area corresponds to 95% confidence interval. Within this interval, the correlation is **not** significant

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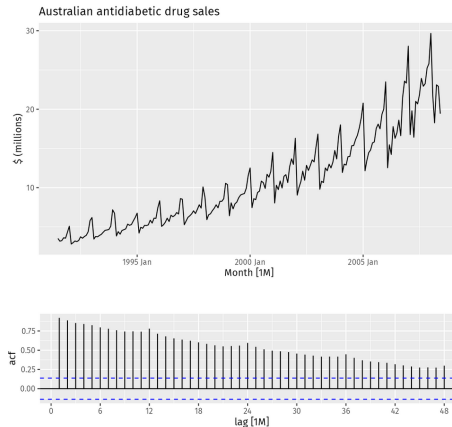
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Interpreting ACF plots

- Trend and Seasonality



- If the ACF plot does not dampen quickly, it is an indication that the time-series is non-stationary

Image from: <https://otexts.com/fpp3/acf.html>

Interpreting ACF plots

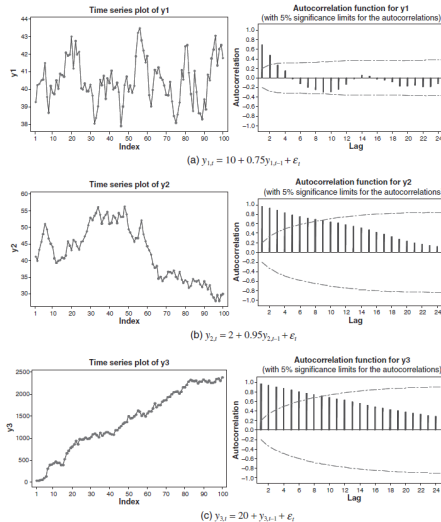


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Playtime!

- Do part 2 of "9. Time Series analysis concepts.ipynb" notebook.
- Expected duration: 30 min

Modeling

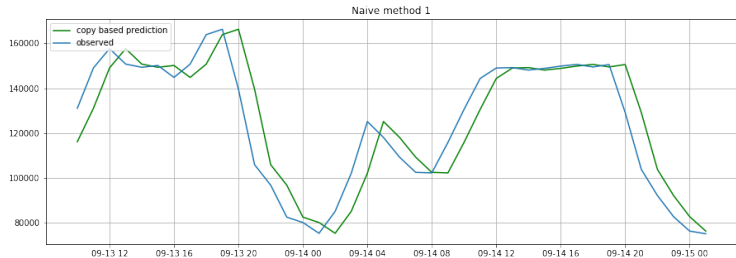
- Naive methods
- Auto Regressive (AR)
- Moving Average (MA)
- ARIMA
- Combination with other models (GP, SVR, MLP, ...)
- LSTM, RNN
- to know more...

Naive methods

- Simplest methods to do forecasting
- Good as a benchmark

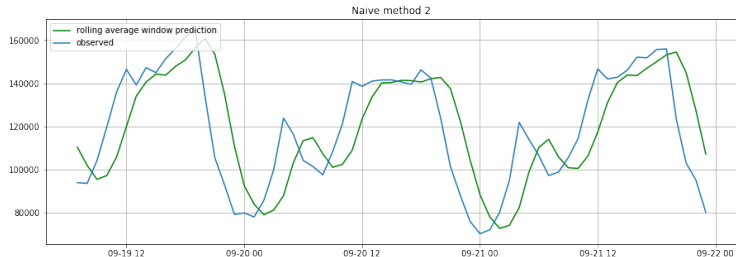
Naive methods

- Simplest methods to do forecasting
- Good as a benchmark
- Two examples
 - Copy last value



Naive methods

- Simplest methods to do forecasting
- Good as a benchmark
- Two examples
 - Copy last value
 - Rolling window (or trivial moving average)



Modeling a stationary time-series

- A stationary time-series can be represented as

$$y(t) = \mu + \sum_{i=0}^{\infty} \psi_i \epsilon_{t-i}$$

where ϵ_t is a white-noise time series with $E[\epsilon_t] = 0$ and

$$\gamma_h = COV(\epsilon_t, \epsilon_{t+h}) = \begin{cases} \sigma^2 & \text{if } h = 0. \\ 0 & \text{if } h \neq 0. \end{cases}$$

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- Above (infinite moving average) serves as a general class of models for any stationary time-series
 - Finite order moving average (MA) models where only a finite number of the ψ_i are non-zero
 - Finite order autoregressive (AR) models where ψ_i are generated using a small number of parameters

Auto Regressive (AR) model

- An Autoregressive model of degree p , $AR(p)$ is defined as (our classical "lags" model):

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

- where p is the auto-regressive lag parameter, $y_{t-1}, y_{t-2}, \dots, y_{t-p}$ are values in previous time periods, $\phi_0, \phi_1, \phi_2, \dots, \phi_p$ are model parameters and ϵ_t the error term.

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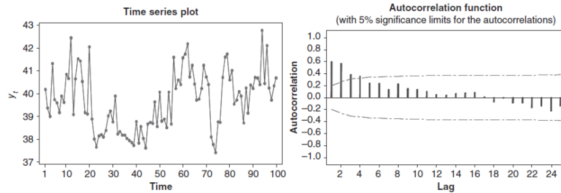
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- Note that one can show that the above is equivalent to the infinite MA form we saw earlier (with a certain structure for the ψ_i s)
- Model estimation via OLS (consistent under stationarity assumptions, biased for small samples), MLE, Yule-walker equations
- p is a *hyperparameter*:
 - An analysis of the PACF (next slide)
 - Trial and error, grid search, optimization...

Auto Regressive (AR) model - ACF Plots

- A realization of the AR(2) process showing exponential decay:

$$y_t = 4 + 0.4y_{t-1} + 0.5y_{t-2} + \epsilon_t$$



- A realization of the AR(2) process showing damped sinusoid behavior:

$$y_t = 4 + 0.8y_{t-1} - 0.5y_{t-2} + \epsilon_t$$

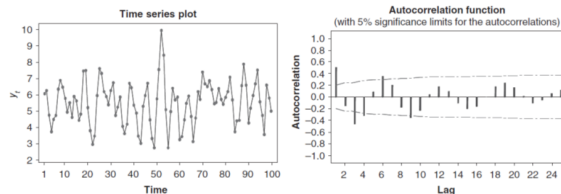
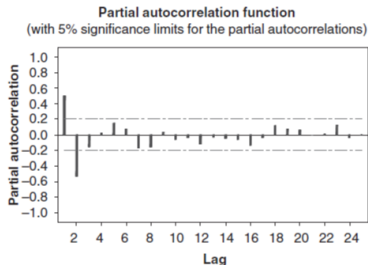
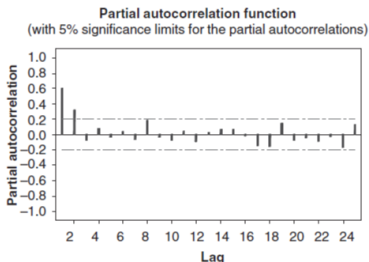


Image from: Montgomery, Douglas C., Cheryl L. Jennings, and Murat Kulahci. Introduction to time series analysis and forecasting. John Wiley & Sons, 2015.

Auto Regressive (AR) model - PACF Plots

- A realization of the AR(2) processes: (a) $y_t = 4 + 0.4y_{t-1} + 0.5y_{t-2} + \epsilon_t$ (b) $y_t = 4 + 0.8y_{t-1} - 0.5y_{t-2} + \epsilon_t$



- The sample PACF plot gives a sense of the number of lags p to consider in an AR(p) model

Image from: Montgomery, Douglas C., Cheryl L. Jennings, and Murat Kulahci. Introduction to time series analysis and forecasting. John Wiley & Sons, 2015.

Moving average (MA) model

- A Moving Average (MA) model of degree q , $MA(q)$ is formulated in terms of previous residuals (forecast errors) as follows ¹

$$y_t = \theta_0 + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \cdots + \theta_q\epsilon_{t-q} + \epsilon_t$$

where q is the moving average trend parameter, $\epsilon_{t-1}, \epsilon_{t-2}, \dots, \epsilon_{t-q}$ are residuals in previous time periods, and ϵ_t the error term.

¹**note:** They are different from “just” the plain moving averages mentioned before. Do not confuse them!

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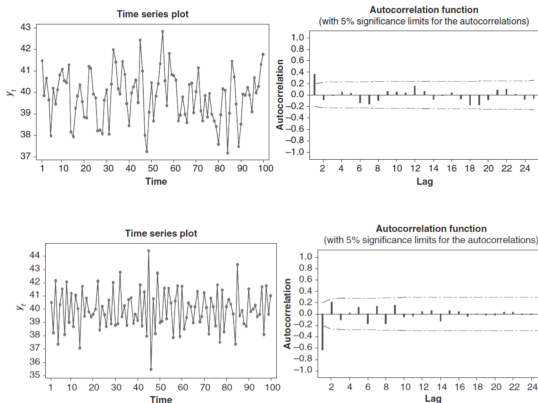
where q is the moving average trend parameter, $\epsilon_{t-1}, \epsilon_{t-2}, \dots, \epsilon_{t-q}$ are residuals in previous time periods, and ϵ_t the error term.

- Model estimation is trickier (Non-linear least squares, MLE)
- q is a *hyperparameter*:
 - An analysis of the ACF can give you a hint
 - Trial and error, grid search, optimization...

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Moving average (MA) model - ACF Plot

- A realization of the MA(1) processes: (a) $y_t = 40 + 0.8\epsilon_{t-1} + \epsilon_t$ (b) $y_t = 40 - 0.8\epsilon_{t-1} + \epsilon_t$



- Sample ACF plot gives a sense of the number of lags q to consider

Image from: Montgomery, Douglas C., Cheryl L. Jennings, and Murat Kulahci. Introduction to time series analysis and forecasting. John Wiley & Sons, 2015.

Mixed Autoregressive Moving Average model (ARMA)

- Combines AR and MA processes
- An ARMA model $\text{ARMA}(p, q)$ is formulated in terms of previous lags and residuals as follows

$$y_t = \delta + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

- where q is the moving average trend parameter, p is the auto-regressive lag parameter, $y_{t-1}, y_{t-2}, \dots, y_{t-p}$ are values in previous time periods, $\epsilon_{t-1}, \epsilon_{t-2}, \dots, \epsilon_{t-q}$ are residuals in previous time periods, $\delta, \phi_1, \phi_2, \dots, \phi_p, \theta_1, \dots, \theta_q$ are model parameters, and ϵ_t the error term.

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- p and q are *hyperparameters*
 - More complex to determine than in the AR and MA models
 - Both ACF and PACF plots exhibit exponential and/or damped sinusoid patterns
 - Trial and error, grid search, optimization...

Summary - ACF/PACF for Stationary processes

Model	ACF	PACF
$MA(q)$	Cuts off after lag q	Exponential decay and/or damped sinusoid
$AR(p)$	Exponential decay and/or damped sinusoid	Cuts off after lag p
$ARMA(p,q)$	Exponential decay and/or damped sinusoid	Exponential decay and/or damped sinusoid

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- I component (which stands for "Integration") determines the degree of differencing
- You can use the AD Fuller test to determine the degree (typically 0, 1 or 2)
- After differencing with order d, to obtain a new signal y'_t , the ARIMA (p, d, q) model is:

$$y'_t = C + \phi_1 y'_{t-1} + \phi_2 y'_{t-2} + \cdots + \phi_p y'_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

ARIMA - Some considerations and extensions

- add exogenous variables (ARIMAX)

$$y'(t) = C + \beta_1 y'(t-1) + \cdots + \beta_p y'(t-p) + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \mathbf{w}^T \mathbf{x}(t) + \epsilon_t$$

- $\mathbf{x}(t)$ is a vector of exogenous variables, $\mathbf{w}$ are coefficients
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- add seasonal submodel (SARIMA)
 - includes a (P,D,Q) seasonal model that shifts the signal up and down according to a seasonality m
- Vector Auto-Regressive (VAR)
 - multiple parallel correlated time series
 - $y(t)$ is now a vector $\mathbf{y}(t) = \{y_1(t), y_2(t), \dots, y_n(t)\}$, and there may be crossed lags (e.g. $y_1(t) = \beta_1 y_1(t-1) + \beta_2 y_2(t-1) \dots$).

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- Each observation in a time-series (in the training set) can be predicted or forecasted using all previous observations - **fitted values**

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- Residuals: difference between observation and fitted value

$$e_t = y_t - \hat{y}_{t|t-1}$$

Image from: <https://otexts.com/fpp3/accuracy.html>

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where the training data is $y_1, \dots, y_T$ and the test data is $y_{T+1}, y_{T+2}, \dots$

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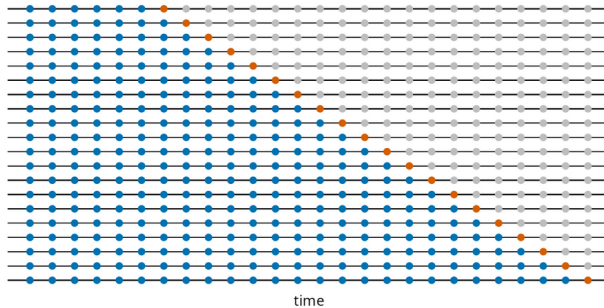
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- *One-step forecasts on the test data*: common way to measure forecast accuracy
 - Here, we use the training data to estimate the model parameters, then apply the model to generate one-step forecasts on the test set (meaning we use all the observed data preceding each test observation)

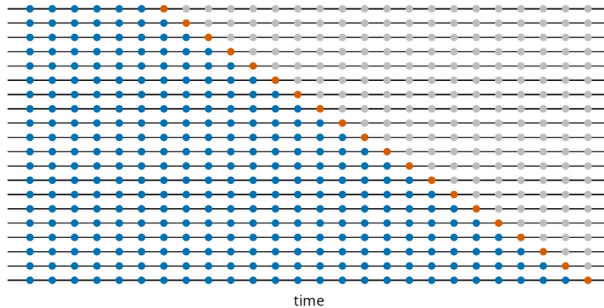
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Time-series cross-validation



- An alternative to the previous definition of training/test sets
- Series of test sets each consisting of a single observation (corresponding training set includes all observations prior to the test observation)

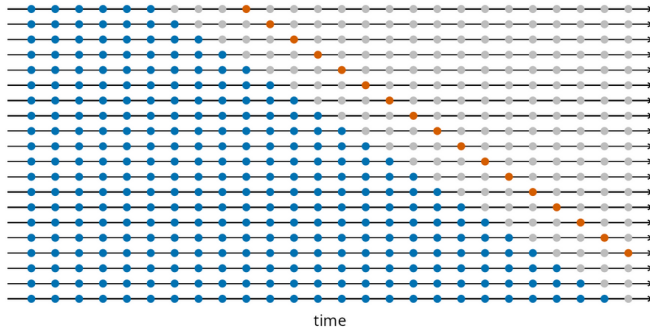
Time-series cross-validation



- An alternative to the previous definition of training/test sets
- Series of test sets each consisting of a single observation (corresponding training set includes all observations prior to the test observation)
- Forecasts are one-step forecasts
- Forecast accuracy is computed by averaging over all test observations

Image from: <https://otexts.com/fpp3/tscv.html>

Time-series cross-validation



- May be also interested in using a multi-step forecast to measure forecast accuracy

Image from: <https://otexts.com/fpp3/tscv.html>

Extending other models with AR lags and differencing

- Sklearn family of Regression
 - SVM, KNN, Random Forest, Multilayer Perceptron, Gaussian Processes, Linear Regression, ...
- Tips:
 - Use PACF to determine AR degree
 - Stationarity is not fundamental, but try detrending (e.g. differencing, subtract a simpler linear model)
 - If you find seasonality cycles, include them as AR terms (e.g. weekly cycle $\rightarrow$ include term for $y_{t-7days}$)
 - If you want to use exogenous variables, go for non-parametric models (e.g. KNN, GP, SVM)

Playtime!

- Do parts 1 to 3 of "9. Time Series analysis modeling.ipynb" notebook.
- Expected duration: 1 hour

References

- Hyndman, R. J. and Athanasopoulos, G. (2018). Forecasting: principles and practice. OTexts. <https://otexts.com/fpp3/>
- Montgomery, Douglas C., Cheryl L. Jennings, and Murat Kulahci. Introduction to time series analysis and forecasting. John Wiley & Sons, 2015.